When Do Amplitude-Encoding Quantum Kernels Help Security Anomaly Detection? A Matched-Scale Study on QKD Telemetry and Network Intrusion Data

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***Abstract*—Amplitude encoding is the most common way to load data into quantum kernels proposed for security tasks. Its fidelity kernel equals the squared cosine similarity of the inputs. It can therefore be computed classically at any scale and compared fairly with classical kernels. We do this on four datasets in two domains. The first is QKD telemetry from a decoy-state BB84 simulator with operating drift. The second is network flows from a synthetic zero-trust simulator, UNSW-NB15 and CIC-IDS2017. Every model is trained on the same data at every training size. In distribution, the amplitude-encoding SVM never beats an RBF-kernel SVM and needs more data at small n. Tree ensembles and neural networks lead. On attack types held out from training the result depends on the data. In both controlled settings the kernel fails, because unseen attacks differ from normal traffic mainly in magnitude and the kernel ignores magnitude. A one-coordinate norm anchor repairs this. On the two public benchmarks it does not fail. A simple norm statistic does not predict which case applies. Held-out attack tests on real traffic are therefore needed. We also find that a Gaussian-process posterior gives worse alert deferral than an isotonic-calibrated random forest, and that drift-free QKD simulators are trivially separable.**

***Index Terms*—Quantum machine learning, quantum kernels, amplitude encoding, intrusion detection, quantum key distribution, zero-day attacks, calibration**

I. Introduction

QUANTUM kernel methods map inputs to quantum states and use state overlaps as a kernel [1], [2]. They are being proposed for security tasks such as intrusion detection [3] and botnet detection [4].

Two facts make such results hard to read. Whether a quantum kernel helps depends on how its geometry matches the data [5]. Proven advantages exist only for specially built problems [6]. Quantum models are also often trained on fewer samples than the baselines, because circuit simulation is slow.

We study amplitude encoding, the most common encoding in this literature. Its kernel is classically computable (Proposition 1). So we can train it at any size, on the same data as classical kernels. Our question does not depend on quantum hardware. Is its inductive bias useful for security anomaly detection, and when?

We answer it in two domains. The first is QKD telemetry, where implementation attacks [7]–[10] perturb channel statistics. The second is flow monitoring for zero-trust networks [11], [12], on a simulator and on UNSW-NB15 and CIC-IDS2017 [13], [14].

Our contributions are as follows.

1) A matched-scale comparison across four datasets in two domains.
2) A zero-day failure of the kernel in both controlled settings, traced to its norm invariance by interventions.
3) A one-coordinate norm anchor that repairs the failure without tuning on held-out attacks.
4) Evidence that the failure does not occur on two public benchmarks, and that a simple statistic cannot predict it.
5) Evidence that a GP posterior is a worse basis for alert deferral than an isotonic-calibrated random forest.

II. Background

Quantum kernels. A quantum kernel evaluates $k(\mathbf{x}, \mathbf{y}) = |\langle\psi(\mathbf{x})|\psi(\mathbf{y})\rangle|^2$ for a data-dependent state [1]. Entangling feature maps can be hard to evaluate classically [6]. Even then they need not beat classical learners on real data [5]. Amplitude encoding is the easy case. Any benefit it gives must come from its inductive bias. Nyström methods [15] and random features [16] let any such kernel scale. Lu et al. study adversarial attacks on quantum classifiers [17].

QKD attack detection. Defenses include hardware monitoring of detector statistics [18] and protocol changes such as decoy states [19] and measurement-device-independent QKD [20]. Mao et al. detect attacks on continuous-variable QKD with neural networks [21]. Ren et al. use machine learning to choose QKD protocols [22]. These studies run on simulated telemetry.

Intrusion detection. Machine-learning intrusion detection is surveyed by Buczak and Guven [23]. Sommer and Paxson warn that detectors often fail on attacks outside their training set [24]. This is why we test held-out attack types. Kitsune popularized autoencoders that never see attack labels [25].

Calibration and robustness. Operators act on alert confidence, so calibration matters [26]. Gaussian processes [27] and other Bayesian models [28] give model-based uncertainty. Platt scaling [29] and isotonic regression [30] recalibrate other models. Randomized smoothing [31] certifies ℓ_2 robustness.

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TABLE I
DATASETS. PUBLIC DATA USES 20,000 TRAINING AND 50,000 TEST FLOWS PER SEED.

Dataset	Samples	Features	Attack types	Seeds
QKD telemetry	2,000	8	4	10
Synthetic flows	8,000	64	5	10
UNSW-NB15	257,673	68	9	5
CIC-IDS2017	2,520,730	69	6	5

III. THE AMPLITUDE-ENCODING KERNEL

Amplitude encoding pads $\mathbf{x} \in \mathbb{R}^d$ with zeros to 2^q entries, with $q = \lceil \log_2 d \rceil$. It then normalizes the result into the state $|\psi(\mathbf{x})\rangle = \sum_i \hat{x}_i |i\rangle$.

Proposition 1. *The fidelity kernel of amplitude encoding is*

$$K_A(\mathbf{x}, \mathbf{y}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{y}) \rangle|^2 = \frac{(\mathbf{x}^\mathsf{T}\mathbf{y})^2}{\|\mathbf{x}\|^2 \, \|\mathbf{y}\|^2}. \tag{1}$$

It costs $\mathcal{O}(d)$ per pair.

Proof. The amplitudes are real and the padding is zero. So the overlap equals $\hat{\mathbf{x}}^\mathsf{T}\hat{\mathbf{y}}$. □

Corollary 1. *$K_A(c\,\mathbf{x}, \mathbf{y}) = K_A(\mathbf{x}, \mathbf{y})$ for every $c \neq 0$. A classifier built on K_A gives $\mathbf{x}$, $2\mathbf{x}$ and $-\mathbf{x}$ the same score.*

On standardized features the origin is the training mean. So the kernel cannot use how far an input lies from typical data. Corollary 1 predicts that K_A helps when labels depend on direction alone. It predicts harm when an attack stands out mainly by being far from normal traffic.

Norm anchor. We append a constant before encoding,

$$\phi_c(\mathbf{x}) = \frac{(\mathbf{x},\, c)}{\sqrt{\|\mathbf{x}\|^2 + c^2}}, \qquad c = \mathrm{median}_{\mathrm{train}} \, \|\mathbf{x}\|. \tag{2}$$

The last amplitude falls as $\|\mathbf{x}\|$ grows. The sign ambiguity disappears. The median rule sets c from training data alone. It costs one extra amplitude.

Sensitivity. The Jacobian of $\mathbf{x} \mapsto \hat{\mathbf{x}}$ has spectral norm exactly $1/\|\mathbf{x}\|$. Normalization shrinks perturbations of large inputs and magnifies those near the origin. It gives no uniform robustness guarantee.

Implementation check. We compare Eq. (1) with a PennyLane [32] simulation of the overlap circuit. They agree to 7×10^{-15}. The circuit takes 26 ms per kernel entry and the closed form 3×10^{-4} ms, about 8×10^4 times faster. Training sizes of a few dozen samples reported for such kernels reflect circuit simulation, not the kernel.

IV. DATA AND PROTOCOL

QKD telemetry. Each sample is one key-generation block of a decoy-state BB84 link over 50 km of fiber [19]. Normal operation drifts in loss, efficiency, misalignment, dark counts and background light. The error-rate estimates carry finite-size noise. Four attacks perturb seven measurements. Each perturbation is scaled by a strength s. We report $s = 1$ and the harder $s = 0.3$. Features are the seven measurements plus a robust Mahalanobis distance [33] to normal blocks. Without drift, a single feature separated attacks perfectly in our first version. We release leakage tests with the simulator.

Synthetic flows. Flow vectors follow NetFlow-like marginals coupled by a Gaussian copula. Five attacks each shift 6 to 7 latent dimensions.

Public data. UNSW-NB15 uses its official split. CIC-IDS2017 is cleaned of missing values, infinite values and 308,381 duplicates. Classes under 100 flows are dropped and the rest form six families. Features pass through $\mathrm{sign}(x)\log(1 + |x|)$.

Models. AE-SVM uses K_A. AE-SVM (anchored) uses K_A on ϕ_c. AE-SVM (nonneg) scales inputs to $[0, 1]$ first. Classical kernels are RBF, Laplacian and cubic polynomial, all in the same SVM. Above 1,500 samples we use a Nyström approximation. Other models are random forests (RF) [34], XGBoost [35] and an MLP. On public data we add a deep network (DNN) and a benign-only autoencoder [25]. On QKD data we add a Laplace GP, a GP and AE-SVM stacking model (Hybrid) [36], isotonic RF and three novelty detectors.

Protocol. Every model in a comparison is trained on the same subsample at every size. Zero-day tests train on all but one attack type and test on the held-out type. We report ROC-AUC there, since it needs no threshold tuned on known attacks. Intervals are 95% confidence intervals over seeds.

V. RESULTS

A. In-distribution detection

AE-SVM never beats RBF-SVM (Table II). On QKD data their MCCs are 0.814 and 0.810, and McNemar finds no significant seed. On synthetic flows their accuracies are 0.947 and 0.948 ($p = 0.54$). On UNSW-NB15 they are 0.862 and 0.863 ($p = 0.24$). On CIC-IDS2017 RBF-SVM leads by 0.001 ($p = 0.001$).

Tree ensembles are strongest overall. RF leads on QKD data and XGBoost on CIC-IDS2017. The DNN has the best F1 on UNSW-NB15. The benign-only autoencoder trails the supervised models in distribution.

The anchor costs at most 0.003 accuracy on public data. On synthetic flows it helps, reaching 0.960 accuracy and 0.983 AUC ($p < 10^{-4}$ against both AE-SVM and RBF-SVM).

B. Training size

No training size favours the amplitude kernel (Fig. 1). On QKD data it trails RBF by 0.03 to 0.04 MCC for $n = 50$ to 200. It catches up only at $n = 1000$. On synthetic flows it trails by 3.6 points at $n = 50$. On CIC-IDS2017 it trails at small n. On UNSW-NB15 it tracks RBF. The Laplacian kernel is the strongest kernel on public data. RF learns fastest on QKD data.

C. Unseen attack types

Table III and Fig. 2 hold the central result. On synthetic flows plain AE-SVM fails. Every held-out type scores between 0.45 and 0.61.

QKD data first looks like a counter-example. There AE-SVM does well. But the QKD features include a Mahalanobis

TABLE II
IN-DISTRIBUTION DETECTION (MEAN ± 95% CI). QKD USES $s = 0.3$ AND $n = 1400$. BOLD MARKS THE BEST MEAN PER COLUMN. A DASH MEANS NOT RUN.

	QKD telemetry		Synthetic flows		UNSW-NB15		CIC-IDS2017	
Model	MCC	AUC	Accuracy	AUC	F1	AUC	F1	AUC
AE-SVM	.814±.023	**.970±.005**	.947±.002	.966±.003	.887±.003	.973±.001	.979±.001	.999±.000
AE-SVM (anchored)	.816±.019	.965±.007	.960±.003	.983±.002	.884±.003	.972±.001	.977±.001	.998±.000
RBF-SVM	.810±.015	.957±.007	.948±.003	.976±.003	.887±.003	.973±.001	.983±.001	.999±.000
Laplacian-SVM	–	–	.946±.003	.976±.002	.899±.004	.978±.001	.987±.001	**.999±.000**
Poly-SVM	–	–	.959±.003	.979±.003	.889±.003	.975±.001	.989±.001	.999±.000
GP	.801±.016	.956±.007	–	–	–	–	–	–
Hybrid	.833±.016	.969±.006	–	–	–	–	–	–
RF	.834±.013	.969±.006	.943±.003	.976±.002	.889±.003	.980±.001	.990±.002	**.999±.000**
RF-isotonic	**.837±.013**	.967±.006	–	–	–	–	–	–
XGBoost	–	–	.962±.003	**.985±.002**	.892±.003	**.981±.001**	**.994±.001**	**1.000±.000**
MLP	–	–	**.965±.001**	.984±.002	.889±.004	.975±.001	.991±.001	**.999±.000**
DNN	–	–	–	–	**.909±.006**	.977±.002	.989±.002	**.999±.000**
Autoencoder	–	–	–	–	.394±.187	.887±.009	.546±.058	.948±.012

TABLE III
MEAN ROC-AUC ON HELD-OUT ATTACK TYPES. "NO MAHAL." REMOVES THE ONLY QKD FEATURE THAT CARRIES MAGNITUDE. BOLD MARKS THE BEST PER COLUMN. A DASH MEANS NOT RUN.

Model	QKD ($s = 0.3$)	QKD, no Mahal.	Synthetic flows	UNSW-NB15	CIC-IDS2017
AE-SVM	0.952	0.554	0.526	0.945	0.715
AE-SVM (nonneg)	–	–	0.593	0.935	0.663
AE-SVM (anchored)	0.949	0.939	0.793	0.940	0.733
RBF-SVM	0.917	**0.940**	**0.839**	0.932	0.658
GP	0.941	–	–	–	–
RF	**0.960**	–	–	0.949	0.820
XGBoost	–	–	–	0.960	0.852
DNN	–	–	–	**0.963**	0.694
Autoencoder	–	–	–	0.860	**0.874**

TABLE IV
SYNTHETIC FLOWS. MEAN HELD-OUT AUC AGAINST THE ANCHOR CONSTANT c (5 SEEDS). $c = 0$ IS PLAIN ENCODING. THE MEDIAN RULE GIVES $c \approx 6.4$.

c	0	1	2	4	8	16	32	median
AUC	0.526	0.549	0.691	0.809	0.771	0.650	0.521	0.793

TABLE V
ACCURACY ON CONTROL TASKS IN $\mathbb{R}^8$ WITH HEAVY-TAILED NORMS (10 SEEDS).

Label depends on	n	AE-SVM	RBF-SVM	GP	RF
Direction only	25	0.707	0.556	0.618	0.606
	100	0.856	0.583	0.652	0.667
	1000	0.966	0.744	0.765	0.794
Norm only	25	0.496	0.799	0.691	0.862
	100	0.499	0.855	0.791	0.924
	1000	0.504	0.958	0.950	0.953

distance, which is a magnitude. Removing it drops AE-SVM to 0.554. RBF-SVM barely changes.

The public benchmarks do not show the failure. AE-SVM matches or beats RBF-SVM on average. On CIC-IDS2017 most supervised models struggle on Bot and PortScan. The benign-only autoencoder is the strongest detector there. This matches the closed-world warning of Sommer and Paxson [24].

D. Mechanism

Corollary 1 offers two causes, sign ambiguity and norm invariance. Interventions separate them (Table III). Scaling to $[0, 1]$ removes the sign ambiguity but keeps norm invariance. It helps little on synthetic flows. The anchor restores the norm. It closes most of the gap on synthetic flows and all of it on QKD data without the Mahalanobis feature. On public data it changes little.

Table IV varies c. Small values leave the norm invisible. Large values squeeze all inputs toward one point. The median rule lands near the best value without seeing held-out attacks.

Control tasks confirm the bias (Table V). When labels depend on direction only, AE-SVM learns from far fewer samples than any baseline. When labels depend on norm only, it stays at chance.

Can one tell in advance which case a dataset is in? We tried the obvious statistic (Table VI). The norm separates attacks best on synthetic flows, where AE-SVM fails. But across held-out splits it does not predict the gap ($\rho = -0.09$, $p = 0.65$, 28 splits). It also misses the QKD ablation. There, removing one coordinate barely changes the norm's power yet flips the outcome. Which features carry magnitude matters more than the norm's overall separating power.

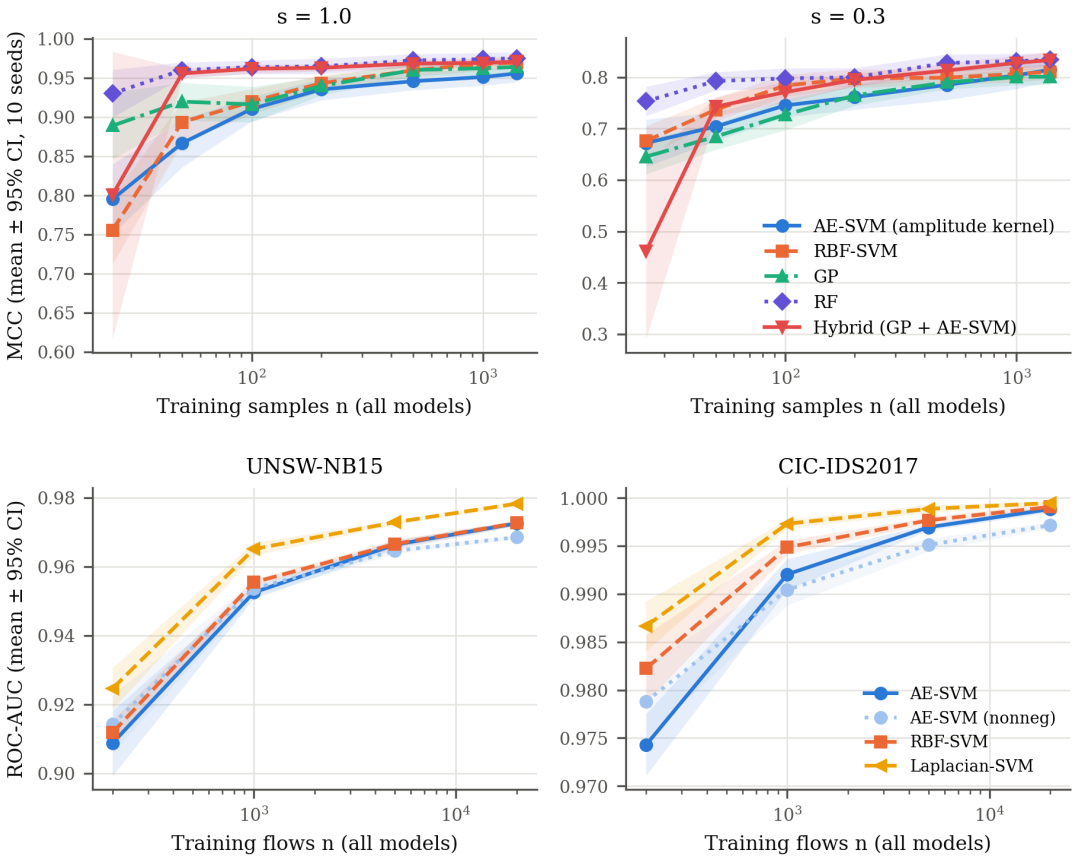


Fig. 1. Every model trained on the same subsample at each n (mean $\pm$ 95% CI). Top, QKD telemetry (MCC). Bottom, public data (ROC-AUC).

TABLE VI
NORM-VERSUS-DIRECTION DIAGNOSTIC, AVERAGED OVER HELD-OUT ATTACK TYPES. NORM AUC SCORES $\|\mathbf{x}\|$ ALONE. DIRECTION AUC SCORES A DIRECTION-ONLY DISTANCE. THE LAST COLUMN IS AE-SVM MINUS RBF-SVM HELD-OUT AUC.

Domain	Norm AUC	Direction AUC	Gap
QKD	0.76	0.58	+0.034
QKD, no Mahal.	0.76	0.59	−0.386
Synthetic flows	0.90	0.66	−0.313
UNSW-NB15	0.51	0.47	+0.013
CIC-IDS2017	0.64	0.67	+0.057

TABLE VII
KERNEL-TARGET ALIGNMENT [37], AVERAGED OVER ATTACK TYPES.

Kernel	QKD	Synthetic	UNSW-NB15	CIC-IDS2017
AE	0.17	0.13	0.38	0.62
RBF	0.48	0.21	0.52	0.67

Kernel-target alignment is lower for K_A on every dataset (Table VII). It predicts the in-distribution ranking. It does not predict held-out behaviour.

E. Calibration and deferral

A GP posterior is often proposed for security monitoring. On QKD data the Laplace GP is the least calibrated supervised model, with ECE 0.047 ± 0.006. Isotonic RF is the best at 0.030. The GP's probabilities crowd the decision boundary (Table VIII). It keeps fewer blocks than isotonic RF or AE-SVM at the same accuracy. At $u < 0.1$ it keeps none. Its uncertainty also barely rises under a gradient-free evasion, from 0.30 to 0.33.

TABLE VIII
QKD TELEMETRY, $s = 0.3$. DEFERRAL BY UNCERTAINTY $u = \sqrt{p(1-p)}$. COVERAGE IS THE SHARE OF BLOCKS KEPT.

	$u < 0.3$		$u < 0.1$	
Model	Coverage	Accuracy	Coverage	Accuracy
GP	69%	97.6%	0%	–
RF-isotonic	80%	97.6%	35%	98.8%
AE-SVM	79%	97.8%	31%	99.3%

F. Robustness and cost

Randomized smoothing gives similar certified radii for AE-SVM and RBF-SVM on synthetic flows, 0.508 and 0.518 on average. AE-SVM abstains on 1.8% of points and RBF-SVM on none. This fits the sensitivity result of Section III.

With 50 Nyström landmarks AE-SVM costs 0.0048 ms per synthetic flow, about 1.2 times faster than RBF-SVM. With 1000 landmarks on public data every kernel SVM is slower than XGBoost, the MLP and the autoencoder. All models stay

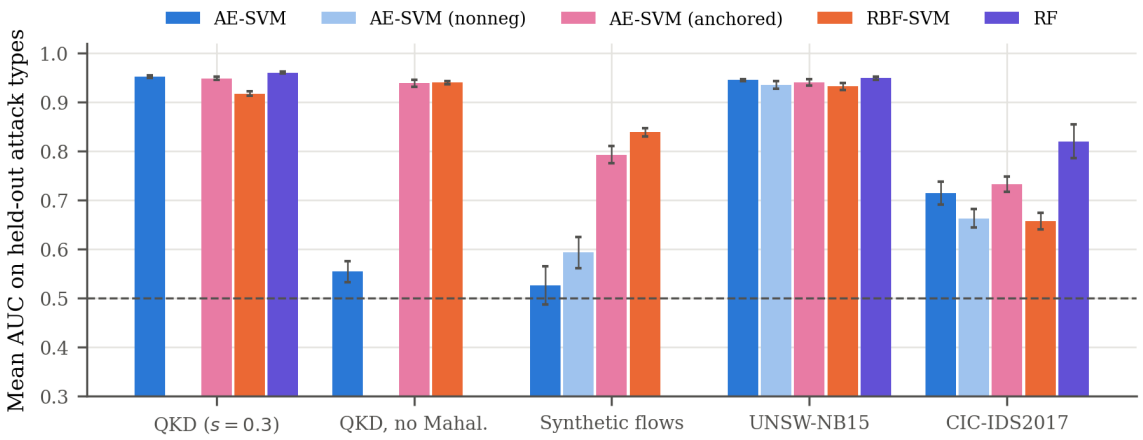


Fig. 2. Mean ROC-AUC on held-out attack types across domains (mean ± 95% CI over seeds).

far below a per-flow budget. Our timings come from a shared machine and show ordering only.

VI. Discussion

Meaning. Amplitude encoding gives no in-distribution gain over an RBF kernel on any of four datasets. At small n it needs more data. Its defining property is an invariance. It sees direction and ignores magnitude. That helps when magnitude is noise. It hurts when unseen attacks differ from normal traffic mainly in size, as in both controlled settings. On real traffic, attack families also differ in which features are active. There the weakness did not appear.

Nothing here bears on quantum speed-ups, since K_A is classically easy. The invariance argument applies to any fidelity kernel whose state preparation normalizes its input. Simulators alone would have led us to the wrong general conclusion. The public benchmarks reversed it.

Recommendations. We suggest five practices for testing quantum or quantum-style detectors.

1) Compare against classical kernels trained on the same data at every n.
2) Report held-out attack results on real traffic.
3) Check what the kernel is invariant to, and whether a feature restores it.
4) If amplitude encoding is used, anchor the norm. In our data it fixed every failure and cost at most 0.005 AUC elsewhere.
5) For simulated telemetry, report single-feature separability and a strength sweep.

Threats to validity. QKD attack sizes are modeling assumptions, and no public QKD attack telemetry exists to our knowledge. The flow simulator is not packet-level traffic. Both public datasets have known labelling artifacts. We subsample them to 20,000 training flows, which favours kernel methods. Kernel settings were not tuned per dataset for any kernel. Tuning might change small gaps but is unlikely to close the 0.3 AUC zero-day gaps. Holding out one attack family only approximates a truly new attack. Certified radii cover ℓ_2 noise, not adaptive evasion.

VII. Conclusion

We tested the amplitude-encoding kernel exactly, at every training size, against classical kernels on the same data. It never beats an RBF kernel in distribution. It needs more data at small n. In both controlled settings it fails on unseen attacks, because it discards input norm. A one-coordinate norm anchor repairs this. On both public benchmarks it does not fail. Quantum detectors should be judged on held-out attack families from real traffic, with their invariances checked first.

Code and Data Availability

Code, both simulators, dataset preparation scripts and all result files are at https://github.com/arnavk23/quasar-security.

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